

KOTELNIKOVO, Russian Federation

WMO: 346550

Lat: 47.633N

Lon: 43.150E

Elev: 154

StdP: 14.61

Time Zone: 4.00 (E04)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.1	1.7	-9.3	3.4	-4.0	-3.2	4.7	3.1	24.0	28.1	21.1	31.9	4.0	70	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	94.3	66.9	91.3	66.0	88.4	65.6	71.4	86.0	69.8	83.7	68.4	81.6	8.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	100.2	75.7	65.4	94.2	74.7	63.8	89.0	73.9	35.2	86.4	33.9	83.2	32.7	81.8	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.5	15.6	DB	-7.2	99.5	6.8	3.2	-12.1	101.8	-16.0	103.7	-19.8	105.4	-24.7	107.8	(4)
(5)			WB	-7.6	73.6	6.6	4.0	-12.4	76.5	-16.3	78.9	-20.0	81.2	-24.8	84.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	24.9	24.8	35.2	50.2	62.3	71.0	74.5	73.0	61.7	48.7	35.0	28.2	(6)
(7)		DBStd	20.16	10.66	10.80	9.06	8.32	7.37	6.27	4.81	5.92	7.12	8.67	9.96	9.68	(7)
(8)		HDD50	3313	779	705	461	100	6	0	0	0	5	127	455	675	(8)
(9)		HDD65	6598	1244	1125	925	446	141	16	0	8	145	507	901	1140	(9)
(10)		CDD50	3043	0	0	2	104	386	630	761	715	354	88	3	0	(10)
(11)		CDD65	855	0	0	0	1	56	196	296	258	45	3	0	0	(11)
(12)		CDH74	9340	0	0	0	31	700	1999	3195	2883	491	41	0	0	(12)
(13)		CDH80	3812	0	0	0	2	216	835	1380	1256	121	2	0	0	(13)
(14)	Wind	WSAvg	7.1	8.6	7.8	8.5	7.4	6.5	6.4	5.6	5.8	6.3	6.4	7.5	8.4	(14)
(15)	Precipitation	PrecAvg	14.3	1.2	0.7	1.0	1.1	1.5	1.7	1.2	1.1	1.1	1.0	1.1	1.3	(15)
(16)		PrecMax	19.2	2.4	1.3	2.8	2.2	3.0	3.6	2.8	2.4	3.9	4.0	2.6	3.0	(16)
(17)		PrecMin	8.6	0.2	0.2	0.1	0.2	0.3	0.5	0.4	0.2	0.1	0.2	0.2	0.4	(17)
(18)		PrecStd	2.5	0.6	0.3	0.7	0.6	0.9	1.0	0.8	0.6	0.9	0.9	0.8	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	49.5	61.2	78.3	91.2	96.3	98.5	98.7	87.6	78.9	60.6	50.2	(19)
(20)			MCWB	44.9	45.6	50.0	55.6	61.0	66.0	70.5	66.3	63.4	61.0	53.0	47.4	(20)
(21)		2%	DB	41.5	42.9	55.6	72.5	85.2	92.5	94.3	93.9	83.4	72.3	55.0	45.5	(21)
(22)			MCWB	39.8	39.8	46.0	55.5	60.6	66.6	68.7	65.7	63.1	58.7	51.1	43.0	(22)
(23)		5%	DB	38.4	39.2	51.4	69.1	81.1	88.8	91.1	90.7	79.9	65.9	50.9	42.6	(23)
(24)			MCWB	36.9	36.8	44.0	54.2	60.3	66.0	66.6	65.8	61.5	56.2	48.0	40.6	(24)
(25)		10%	DB	36.1	36.2	47.5	65.4	77.1	85.2	87.9	87.0	76.0	61.4	47.8	39.1	(25)
(26)			MCWB	34.8	34.3	42.3	52.8	59.3	66.0	65.9	65.5	60.0	53.8	45.2	37.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	45.4	51.4	60.0	65.4	74.7	73.8	71.9	69.0	63.4	55.1	48.1	(27)
(28)			MCD B	46.3	49.5	58.2	70.9	76.9	85.9	94.4	85.8	81.3	74.2	59.2	50.1	(28)
(29)		2%	WB	40.0	40.0	47.8	57.4	63.9	71.6	71.1	70.0	65.5	59.7	51.2	43.4	(29)
(30)			MCD B	41.2	42.2	54.2	68.4	76.2	84.8	86.9	84.6	78.0	68.7	54.1	45.3	(30)
(31)		5%	WB	37.2	37.0	44.8	55.3	62.4	69.5	69.8	68.5	63.6	57.4	48.2	40.6	(31)
(32)			MCD B	38.2	38.8	49.9	64.9	76.0	82.1	84.0	82.9	74.6	64.5	50.6	42.4	(32)
(33)		10%	WB	34.9	34.5	42.4	53.6	60.9	67.5	68.4	67.1	61.8	54.9	45.4	37.7	(33)
(34)			MCD B	35.9	35.9	47.4	64.1	74.3	79.5	81.7	81.5	72.1	61.2	47.9	39.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	10.8	13.3	19.7	21.3	20.3	21.4	22.3	21.0	17.9	11.5	8.9	(35)
(36)			MCDBR	9.3	11.7	21.4	26.7	27.8	26.9	26.9	28.5	27.7	25.8	14.7	9.9	(36)
(37)			MCWBR	7.9	9.1	13.5	13.2	10.0	9.8	9.1	9.6	12.7	14.3	11.2	8.4	(37)
(38)		5% WB	MCDBR	9.4	9.9	19.0	22.7	22.3	22.0	21.6	22.7	21.9	20.7	13.4	9.8	(38)
(39)			MCWBR	8.5	8.2	12.8	12.6	9.8	10.4	8.3	9.8	11.4	12.0	10.6	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.374	0.408	0.455	0.426	0.420	0.441	0.443	0.416	0.381	0.353	0.344	(40)	
(41)		taud	2.162	2.083	2.115	2.043	2.219	2.282	2.229	2.200	2.233	2.285	2.266	2.180	(41)	
(42)		Ebn at Noon	221	239	252	252	266	268	260	253	248	237	217	203	(42)	
(43)		Edh at Noon	28	37	42	49	43	41	43	42	37	30	25	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	288	550	932	1472	1887	2063	2020	1803	1313	761	351	211	(44)	
(45)		RadStd	69	98	102	136	179	148	96	119	126	107	61	26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+144

Nomenclature: See separate page